

Executive Project Report

AI Student Impact Analysis

1. Introduction

This report summarizes the comprehensive end-to-end data science lifecycle executed for the AI Student Impact Analysis project. The primary objective was to investigate the influence of Generative AI tools on students' academic performance (GPA), skill retention, and psychological well-being (Burnout Risk). We transformed a raw dataset into a fully functional, highly interactive Cyberpunk-themed Dashboard that empowers educators and researchers to make data-driven decisions.

2. Exploratory Data Analysis (EDA)

In the initial phase, we conducted thorough exploratory analysis to understand the distributions and correlations within our dataset.

- **Univariate Analysis:** We utilized Plotly histograms to analyze the distribution of GPA, traditional study hours, and weekly GenAI hours.
- **Bivariate Analysis:** We mapped the relationship between GenAI usage and its impact on Skill Retention, discovering key thresholds where over-reliance on AI correlates with diminished foundational skills.
- **Visual Design:** All EDA outputs were embedded into the Streamlit dashboard using Custom HTML Glassmorphism tables and neon-styled Plotly charts.

3. Feature Engineering

To maximize the predictive power of our Machine Learning models, we engineered several critical features:

- **GPA_Improvement:** The delta between Post-Semester and Pre-Semester GPA.
- **AI_Usage_Ratio:** The proportion of GenAI hours compared to total study hours.
- **Dependency_per_Hour:** A metric measuring the psychological reliance on AI normalized by usage volume.
- **Burnout_Index:** A composite index formulated from Anxiety Levels and Perceived AI Dependency.

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4. Predictive Modeling (Machine Learning)

We developed three distinct XGBoost models to capture different facets of student outcomes:

- **A. GPA Prediction:** XGBRegressor predicting final GPA based on study habits and AI usage.
- **B. Skill Retention:** XGBRegressor measuring the positive/negative impact of AI on cognitive retention.
- **C. Burnout Risk:** XGBClassifier categorizing students into Low, Medium, or High burnout risk tiers, displaying predictions via glowing Glassmorphism cards.

5. Unsupervised Clustering (Student Personas)

We employed K-Means Clustering to segment the student population into unique behavioral personas. The clusters were analyzed and officially named based on their distinct traits:

1. The AI Pioneers: High GenAI hours, advanced prompt skills, high GPA.
2. The Traditional Scholars: Low AI usage, high traditional study hours.
3. The Balanced Achievers: Optimal mix of AI and traditional studying.
4. The Stressed Adopters: High AI dependency coupled with high anxiety.
5. The Casual Users: Low overall academic engagement.
6. The Tech Enthusiasts: High tool diversity, moderate study hours.
7. The Cautious Learners: Slow to adopt AI, high skill retention.
8. The Hyper-dependent: Severe reliance on AI, resulting in critical burnout risks.

STATUS: PROJECT COMPLETE